

Project Title:

Azure Cost Management Budget and Anomaly Alerting

Project Number: P016

Domain: Cloud Computing

Problem Statement:

Cloud spending can increase before teams notice, especially when resources are created without clear ownership or spending limits. Without budget thresholds, department owners may discover overspending only after the cost has already accumulated. Untagged resources also make it difficult to connect costs to a department, project, or owner. Unexpected spending changes may further require investigation. Our solution uses Azure Cost Management to monitor spending, set budget thresholds, analyze costs, apply ownership tags, and detect unusual cost patterns.

Objectives:

- To create and manage an Azure Cost Management budget for the Azure for Students subscription.
- To configure actual-cost alerts at 80% and 100% of the monthly budget.
- To apply resource tags for department, project, and owner-based cost organization.
- To create a Cost Analysis view grouped by service for monthly cost-driver analysis.
- To configure daily cost anomaly detection by resource group.
- To send email notifications for budget and anomaly conditions.
- To understand practical Azure cost-control and monitoring using student resources and credits.

Use Cases:

- **Budget overrun alerting:** Department owners receive notifications when actual spending reaches the configured budget thresholds.
- **Monthly cost-driver analysis:** Cost Analysis grouped by service can be used to identify the top five cost drivers when cost data is available.
- **Cost anomaly detection:** Unusual daily spending patterns can trigger an anomaly alert for investigation.
- **Resource ownership and allocation:** Tags connect resources with department, project, and owner information for clearer cost organization.
- **Cost investigation:** Teams can use Cost Analysis to investigate the service, resource group, and ownership information behind unusual spending.

Requirements:

- Azure for Students subscription
- Azure Resource Group
- Azure Storage Account or another low-cost Azure resource
- Azure Cost Management
- Monthly budget configuration
- Actual-cost threshold alerts
- Cost Analysis
- Resource tags
- Cost anomaly alert rule
- Email recipients for notifications
- Internet connection / Azure Portal access

Bottlenecks / Challenges:

- Cost Analysis may initially show no cost data for a newly created student subscription.
- Budget alerts notify users but do not automatically stop or delete Azure resources.
- Missing or inconsistent resource tags can make cost allocation and ownership analysis incomplete.
- Student Azure credits are limited, so testing must avoid unnecessary paid resources.
- Cost data may take time to become available before a real top-five cost-driver report can be produced.
- Anomaly alerts identify unusual spending patterns, but the team still needs to investigate the responsible resource or service.

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Live implementation completed: RG-CostManagement, costalertstorage001, resource tags, ■10 monthly budget, 80%/100% actual-cost alerts, Monthly Cost by Service view, and an Active Daily anomaly by resource group rule.